

Reporting Summary Draft

This draft is intended to populate the official Nature Portfolio Reporting Summary if the submission system requests the editable form. It is not a replacement for the official form.

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Provided in the editorial cover letter for double-anonymous peer review.

Statistics

- Exact sample sizes: primary breast S3 residual associations use 157 H&E-WTA paired contours; cervical stress-test associations use 215 H&E-WTA paired contours. Source-data tables report the contour count for each association.
- Measurement unit: one statistical row is one registered spatial-omics contour with matched H&E embedding summaries and WTA program summaries.
- Statistical tests: rank-residualized partial Spearman correlations were used for program-feature associations. Spatial-null empirical P values are two-sided and computed from compartment-aware permutations.
- Covariates: spatial-omics-derived contour label where nonconstant, centroid-position covariates and boundary-distance summaries or bins. In breast S3 analyses the contour label is constant and omitted from the model matrix.
- Multiple testing and discovery framing: the manuscript reports ranked candidate program-feature pairs and treats spatial permutations, block bootstrap, leave-one-block checks, local mismatch controls, centroid jitter, registration perturbation and nested spatial holdout as robustness checks rather than cohort-level validation.

Software and code

- Data collection: no new experimental data collection was performed for this manuscript.
- Data analysis: mTM analysis, direct WSI embedding, contour aggregation, statistical defense, figure composition and packaging are implemented in code identified in the manuscript Code availability section and in the editorial cover letter.
- Code availability: public code and release links are provided in the manuscript Code availability section; the cover letter repeats the identifiers for editorial convenience.

Data

- Source data: CSV figure source data for Fig. 1 and supplementary figures are included in the submission package; separate machine-readable Supplementary Data CSVs are included for robustness tables.
- Referenced datasets: raw 10x Genomics Atera WTA and H&E input files are public/vendor example datasets available at <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-breast-cancer> and <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-cervical-cancer>.
- Restrictions: raw vendor inputs are not redistributed with this submission package. Public/vendor source datasets are identified in the manuscript Data availability statement.

Human research participants and data use

- The analysis uses public/vendor example spatial WTA and H&E data.
- The study did not involve new human sample collection, intervention, recruitment or generation of new identifiable participant data.
- Sex, gender, race, ethnicity and other participant-level characteristics were not analyzed because the public/vendor example datasets do not support participant-level inference for this study.

Life sciences study design

- Sample size rationale: all available paired contours passing the locked contour/WTA/image matching workflow were analyzed for the reported breast S3 and cervical stress-test association sets.

- Data exclusions: contours without matched H&E-WTA summaries or required program/covariate values were not included in the corresponding association table; source-data files expose the analyzed contour counts.
- Replication: cervical Atera WTA is treated as a second epithelial cancer stress test, not a direct replication of breast luminal estrogen-response biology. PLIP and UNI provide model-sensitivity evidence at the program-family level.
- Randomization and blinding: no experimental randomization or blinding was introduced because this was a computational secondary analysis of public/vendor example data. Null distributions were generated by permutation within spatial/contour strata.
- Validation limits: no IHC, protein-validation, clinical deployment or morphology-only diagnostic claim is made.